

Methodology Pipeline: SRF for BME Stabilization

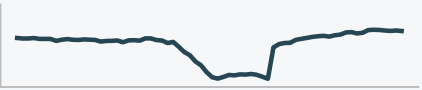
System Model & Theoretical Framework

State Vector

$$\begin{bmatrix} \text{P} & \text{S} & \text{R} \end{bmatrix}^T$$

derived: $\sqrt{\text{Vol}}_t, P_{\text{target}, t}$

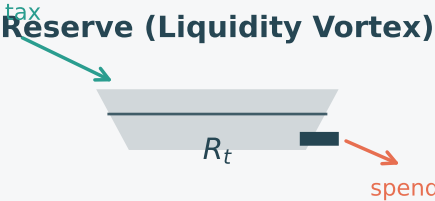
Price SDE



Supply Dynamics



Reserve (Liquidity Vortex)



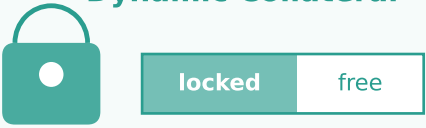
Stage output: Dynamic Differential Equation System $\{dP_t, dS/dt, dR/dt\}$

Closed-Loop Stabilization Engine

PD Controller



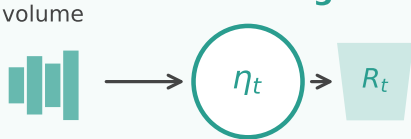
Dynamic Collateral



Safeguards



Self-Financing Tax



Stage output: Control signals $\{\Delta P_{\text{SRF}, t}, \ell_t, \eta_t\}$

Operational Implementation

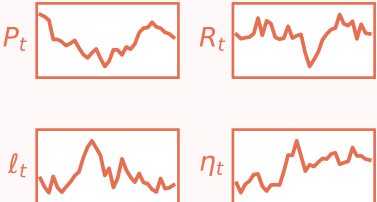
Inputs & Calibration



Daily Integration Loop



Output Trajectories



Stage output: Simulated trajectories $\{P_t, R_t, \ell_t, \eta_t\}_{t=1}^T$